

Loading & Inspecting Data

Before analysis, we need to understand the dataset

Lecture 2 • Module 1: Data Literacy & Python

Press Space to start →

Module 1

Loading & Inspecting Data

2.1 What We Already Know

Last lecture

- Financial data is information about a company over time
- Stock prices are one kind of financial data
- Historical prices are often stored in CSV files

Today's focus

We will not try to predict stock prices yet. We will first learn how to **read**, **inspect**, and **judge** whether a dataset is usable.

Big idea: good analysis starts with good data habits.

2.2 Learning Objectives

By the end of this lecture, students can...

- Import Pandas
- Read a CSV file into a DataFrame
- View the first and last rows
- Identify the columns and data types

And also...

- Recognize time-series data
- Notice missing values and duplicates
- Ask whether the dataset is trustworthy
- Decide what needs cleaning later

2.3 The Data Workflow

- 1 **Read the file** with Pandas
- 2 **Inspect the data** from several angles
- 3 **Find problems** before analysis
- 4 **Clean later** when we know what needs fixing

Today is about steps 1 and 2.

2.4 Why Inspect Data First?

If we inspect first...

- we notice missing values
- we understand the column meaning
- we catch strange dates or duplicate rows
- we avoid misleading conclusions

If we skip inspection...

- charts may look wrong
- calculations may be inaccurate
- results may be based on bad data
- decisions become less reliable

2.5 Meet the Dataset

Example dataset

Date

Close

Open

Volume

High

Low

Company example

- Apple (AAPL)
- Tesla (TSLA)
- Google (GOOGL)
- NVIDIA (NVDA)

You can choose a company they already know.

2.6 Import Pandas

Notebook cell

```
import pandas as pd
```

What this means

- Pandas helps us work with tables
- **pd** is a short nickname
- We will use it in almost every lecture

2.7 Read the CSV File

Notebook cell

```
df = pd.read_csv("apple.csv")
```

What happens here?

- the file is loaded into Python
- the result is stored in **df**
- **df** means DataFrame

2.8 What Is a DataFrame?

Think of it as...

- a smart spreadsheet
- a table that Python can understand
- rows = observations
- columns = variables

Why this matters

A DataFrame is the structure we will use for nearly all of our analysis in Module 1.

2.9 First Look: Top and Bottom

The first rows

```
df.head()
```

The last rows

```
df.tail()
```

Use these commands to quickly see whether the dataset looks reasonable.

2.10 What Does the Dataset Look Like?

Date	Open	High	Low	Close	Volume
2024-01-02	248.0	252.1	247.3	251.2	87,000,000

One row usually represents one trading day.

2.11 How Big Is It?

Dataset size

`df.shape`

Example: **(252, 6)**

What the numbers mean

- 252 rows = 252 trading days
- 6 columns = 6 variables

2.12 What Columns Are There?

```
df.columns
```

Date

when the trading happened

Open

first price of the day

High

highest price

Low

lowest price

Close

last price of the day

Volume

shares traded

2.13 What Kind of Data Is Each Column?

Use this command

```
df.info()
```

Why it helps

- tells us the data type
- shows whether values are missing
- helps us spot unexpected text or numbers

2.14 Selecting Columns

Single column

```
df["Close"]
```

Multiple columns

```
df[["Date", "Close"]]
```

We do not always need every column. We keep the information that answers our question.

2.15 What Is Time-Series Data?

Definition

Time-series data is data that changes over time. Each row is tied to a specific date or time.

Examples

- stock prices
- monthly sales
- yearly population
- weather temperature

2.16 One Row = One Trading Day

Date	Close
2024-01-02	251.2
2024-01-03	249.8
2024-01-04	253.1

The order of rows matters because the dates tell a story over time.

2.17 Why Does Order Matter?

Correct order

Jan → Feb → Mar → Apr

Wrong order

Mar → Jan → Apr → Feb

If the dates are not sorted, graphs and conclusions can become misleading.

2.18 A Date Check

Ask these questions

- Is the earliest date at the top or bottom?
- Are the dates in the same format?
- Are there any impossible dates?

Why this matters

A dataset can look fine at first glance, but still contain small issues that change the result of an analysis.

2.19 Missing Values

What is missing data?

A value is missing when the dataset has a blank or unknown entry.

Common causes

- market was closed
- data download error
- incomplete record
- formatting problem

2.20 Finding Missing Values

Notebook cell

```
df.isna().sum()
```

What to look for

- which columns have missing values
- how many values are missing
- whether the missing values matter

2.21 Duplicate Rows

Notebook cell

```
df.duplicated().sum()
```

Why duplicates matter

Duplicate rows can make the dataset look larger than it really is and can distort averages or charts.

2.22 Strange Values

Negative stock price

✗ Almost certainly wrong

Volume = 0

⚠ Could be a holiday or issue

Future date

✗ Probably a data error

Good analysts do not trust a number just because it is in the file.

2.23 What Does `df.info()` Tell Us?

Command

```
df.info()
```

Useful clues

- total rows
- non-missing values
- data type of each column

2.24 What Does `df.describe()` Tell Us?

Command

```
df.describe()
```

Useful clues

- average
- minimum
- maximum
- spread of values

2.25 Mini Detective Checkpoint

Questions:

- How many rows are in the file?
- Which column stores the date?
- Are the rows in order?
- Any missing values?

2.26 Think–Pair–Share

Prompt

If a dataset has missing values and duplicate rows, should we use it right away? Why or why not?

Think

1 minute alone

Pair

2 minutes with a partner

Share

1 answer with the class

2.27 What You Should Remember

Today's wins

-  Read CSV with Pandas
-  Inspect rows and columns
-  Recognize time-series data
-  Notice missing values and duplicates

Most important habit

Do not trust the dataset blindly. Inspect it first.

2.28 Exit Ticket

Write one sentence for each:

1. What is a DataFrame?
2. Why do we check missing values?
3. Why does row order matter in stock data?

Next Lecture

Cleaning Data

Next we will learn how to:

- Handle missing values
- Sort dates correctly
- Fix data types
- Save a clean dataset